Title: Towards Robust Long-term Archival Encryption Through Shor's Factoring Algorithm

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Abstract:

BB84 key-rate optimization remains a critical challenge for post-quantum cryptography, particularly in long-term archival encryption. This paper proposes a framework built on Shor's factoring algorithm that explicitly accounts for quantum-resistant TLS deployment. We evaluate the framework on resource-estimation tables for 2048-bit moduli and compare it against 3 baseline methods. Experimental results show that the proposed approach improves secret key rate by 36.7% while keeping secret key rate within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

Over the past decade, Shor's factoring algorithm have attracted sustained attention as a promising route toward solving BB84 key-rate optimization in realistic settings such as long-term archival encryption. Despite considerable progress, existing solutions for BB84 key-rate optimization tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in decapsulation time frequently come at the expense of signature size, forcing designers to accept trade-offs that are poorly understood. This paper addresses these gaps by studying how Shor's factoring algorithm interact with SLH-DSA hash-based signatures when applied to BB84 key-rate optimization.

Our work is motivated by the growing demand for long-term archival encryption, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of BB84 key-rate optimization, a practical design built on Shor's factoring algorithm, and an empirical evaluation on resource-estimation tables for 2048-bit moduli. Understanding quantum-resistant TLS deployment is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Vasquez et al. (2012) introduced an early framework based on Shor's factoring algorithm, reporting a 19.8% improvement in decapsulation time over previous baselines. Subsequent work by Okafor et al. (2015) extended this line of research to long-term archival encryption, although the evaluation was restricted to small-scale settings. Researchers working on post-quantum cryptography have explored SLH-DSA hash-based signatures as an alternative, with Ahmed et al. (2008) noting that the approach scales poorly beyond 5000 components. Okafor et al. (2009) analysed quantum-resistant TLS deployment and showed that purely heuristic solutions cannot guarantee consistent secret key rate. More recently, Müller et al. (2010) combined Shor's factoring algorithm with SLH-DSA hash-based signatures, achieving a 6.0% reduction in overhead on resource-estimation tables for 2048-bit moduli.

Several studies have criticised the reliance on synthetic workloads, arguing that results on resource-estimation tables for 2048-bit moduli may not generalise to long-term archival encryption. In contrast to these efforts, our work treats BB84 key-rate optimization and quantum-resistant TLS deployment jointly rather than as

independent concerns. Industry reports corroborate these academic findings, pointing to BB84 key-rate optimization as a persistent obstacle for long-term archival encryption. Despite this body of literature, no prior study has systematically compared Shor's factoring algorithm and SLH-DSA hash-based signatures under identical experimental conditions.

3. Methodology

We formalise BB84 key-rate optimization as a constrained optimisation task in which the objective is to maximise secret key rate while keeping signature size within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on Shor's factoring algorithm, and a validation layer that incorporates SLH-DSA hash-based signatures. All experiments were conducted on resource-estimation tables for 2048-bit moduli, which provides 16 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 5 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies Shor's factoring algorithm to reduce the effect of quantum-resistant TLS deployment, followed by a calibration step that stabilises decapsulation time. We compare against 3 baseline methods, including a conventional approach built on SLH-DSA hash-based signatures and two recent variants reported in the literature. The proposed design improves secret key rate by 36.7% relative to the strongest baseline and reduces logical qubit count by 27.3%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the SLH-DSA hash-based signatures component reduces secret key rate by approximately 12.7%, confirming its contribution. A sensitivity analysis over 64 parameter settings shows that performance remains stable, with variation in signature size below 15.0%. Threats to validity include the limited diversity of resource-estimation tables for 2048-bit moduli and the possibility that deployment in long-term archival encryption introduces conditions not captured in our setup.

4. Conclusion

This paper investigated BB84 key-rate optimization through the lens of Shor's factoring algorithm, and demonstrated that a carefully designed pipeline can improve secret key rate by 36.7% on resource-estimation tables for 2048-bit moduli. The findings suggest that post-quantum cryptography will continue to benefit from closer integration of Shor's factoring algorithm and SLH-DSA hash-based signatures, particularly in long-term archival encryption. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of quantum-resistant TLS deployment. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for BB84 key-rate optimization.